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Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced

Paper 3: Statistics and Mechanics

Friday 15 June 2018 – Afternoon

Time: 2 hours

Paper Reference

9MA0/03

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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SECTION B: MECHANICS

Unless otherwise stated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

6. At time t seconds, where $t \geq 0$, a particle P moves in the x - y plane in such a way that its velocity v ms^{-1} is given by

$$\mathbf{v} = t^{-\frac{1}{2}}\mathbf{i} - 4t\mathbf{j}$$

When $t = 1$, P is at the point A and when $t = 4$, P is at the point B .

Find the exact distance AB .

(6)

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



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Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors due east and due north respectively and position vectors are given relative to the fixed point O .]

A particle P moves with constant acceleration.

At time $t = 0$, the particle is at O and is moving with velocity $(2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-1}$

At time $t = 2$ seconds, P is at the point A with position vector $(7\mathbf{i} - 10\mathbf{j})\text{m}$.

- (a) Show that the magnitude of the acceleration of P is 2.5 m s^{-2}

(4)

At the instant when P leaves the point A , the acceleration of P changes so that P now moves with constant acceleration $(4\mathbf{i} + 8.8\mathbf{j})\text{ m s}^{-2}$

At the instant when P reaches the point B , the direction of motion of P is north east.

- (b) Find the time it takes for P to travel from A to B .

(4)



Question 8 continued

Handwriting practice area with 20 horizontal lines.

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Question 8 continued

Lined area for writing the answer to Question 8.

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Question 8 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 8 is 8 marks)



9.

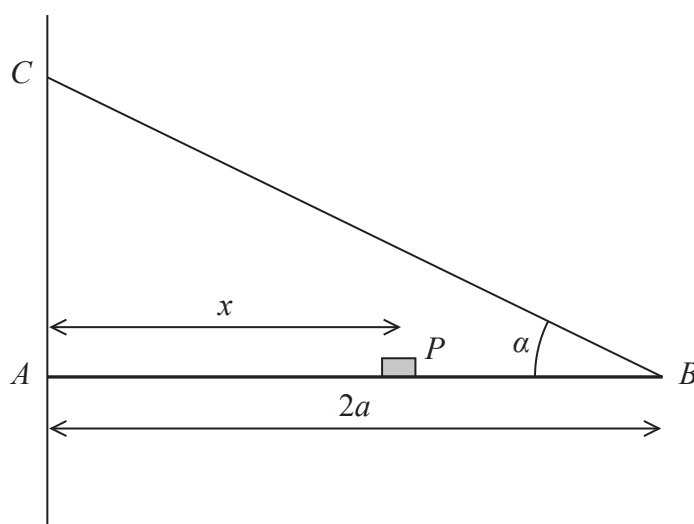


Figure 3

A plank, AB , of mass M and length $2a$, rests with its end A against a rough vertical wall. The plank is held in a horizontal position by a rope. One end of the rope is attached to the plank at B and the other end is attached to the wall at the point C , which is vertically above A .

A small block of mass $3M$ is placed on the plank at the point P , where $AP = x$. The plank is in equilibrium in a vertical plane which is perpendicular to the wall.

The angle between the rope and the plank is α , where $\tan \alpha = \frac{3}{4}$, as shown in Figure 3.

The plank is modelled as a uniform rod, the block is modelled as a particle and the rope is modelled as a light inextensible string.

- (a) Using the model, show that the tension in the rope is $\frac{5Mg(3x + a)}{6a}$ (3)

The magnitude of the horizontal component of the force exerted on the plank at A by the wall is $2Mg$.

- (b) Find x in terms of a . (2)

The force exerted on the plank at A by the wall acts in a direction which makes an angle β with the horizontal.

- (c) Find the value of $\tan \beta$ (5)

The rope will break if the tension in it exceeds $5Mg$.

- (d) Explain how this will restrict the possible positions of P . You must justify your answer carefully. (3)



Question 9 continued

Handwriting practice area with 20 horizontal lines.

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Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 9 is 13 marks)



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Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

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Question 10 continued

Lined area for writing the answer to Question 10.

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 15 marks)

TOTAL FOR SECTION B IS 50 MARKS
TOTAL FOR PAPER IS 100 MARKS

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Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Friday 14 June 2019			
Afternoon		Paper Reference 9MA0-32	
Mathematics			
Advanced			
Paper 32: Mechanics			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 6 marks)



2. A particle, P , moves with constant acceleration $(2\mathbf{i} - 3\mathbf{j})\text{m s}^{-2}$

At time $t = 0$, the particle is at the point A and is moving with velocity $(-\mathbf{i} + 4\mathbf{j})\text{m s}^{-1}$

At time $t = T$ seconds, P is moving in the direction of vector $(3\mathbf{i} - 4\mathbf{j})$

- (a) Find the value of T .

(4)

At time $t = 4$ seconds, P is at the point B .

- (b) Find the distance AB .

(4)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



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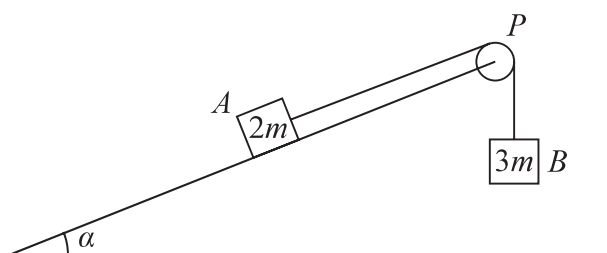


Figure 1

Two blocks, A and B , of masses $2m$ and $3m$ respectively, are attached to the ends of a light string.

Initially A is held at rest on a fixed rough plane.

The plane is inclined at angle α to the horizontal ground, where $\tan \alpha = \frac{5}{12}$

The string passes over a small smooth pulley, P , fixed at the top of the plane.

The part of the string from A to P is parallel to a line of greatest slope of the plane. Block B hangs freely below P , as shown in Figure 1.

The coefficient of friction between A and the plane is $\frac{2}{3}$

The blocks are released from rest with the string taut and A moves up the plane.

The tension in the string immediately after the blocks are released is T .

The blocks are modelled as particles and the string is modelled as being inextensible.

- (a) Show that $T = \frac{12mg}{5}$ (8)

After B reaches the ground, A continues to move up the plane until it comes to rest before reaching P .

- (b) Determine whether A will remain at rest, carefully justifying your answer. (2)
- (c) Suggest two refinements to the model that would make it more realistic. (2)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 12 marks)



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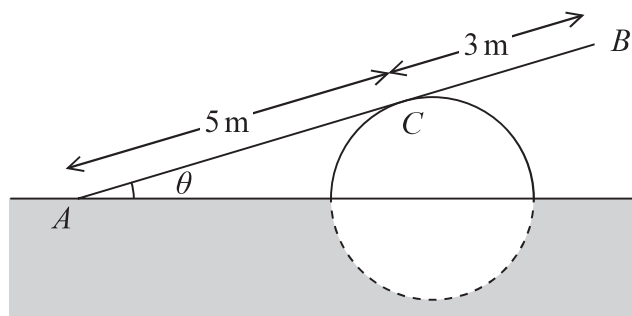


Figure 2

A ramp, AB , of length 8 m and mass 20 kg, rests in equilibrium with the end A on rough horizontal ground.

The ramp rests on a smooth solid cylindrical drum which is partly under the ground. The drum is fixed with its axis at the same horizontal level as A .

The point of contact between the ramp and the drum is C , where $AC = 5$ m, as shown in Figure 2.

The ramp is resting in a vertical plane which is perpendicular to the axis of the drum, at an angle θ to the horizontal, where $\tan \theta = \frac{7}{24}$

The ramp is modelled as a uniform rod.

(a) Explain why the reaction from the drum on the ramp at point C acts in a direction which is perpendicular to the ramp. (1)

(b) Find the magnitude of the resultant force acting on the ramp at A . (9)

The ramp is still in equilibrium in the position shown in Figure 2 but the ramp is not now modelled as being uniform.

Given that the centre of mass of the ramp is assumed to be closer to A than to B ,

(c) state how this would affect the magnitude of the normal reaction between the ramp and the drum at C . (1)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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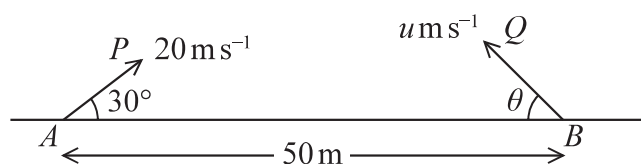


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**Figure 3**

The points A and B lie 50 m apart on horizontal ground.

At time $t = 0$ two small balls, P and Q , are projected in the vertical plane containing AB .

Ball P is projected from A with speed 20 m s^{-1} at 30° to AB .

Ball Q is projected from B with speed $u \text{ m s}^{-1}$ at angle θ to BA , as shown in Figure 3.

At time $t = 2$ seconds, P and Q collide.

Until they collide, the balls are modelled as particles moving freely under gravity.

- (a) Find the velocity of P at the instant before it collides with Q . (6)
- (b) Find
- the size of angle θ ,
 - the value of u . (6)
- (c) State one limitation of the model, other than air resistance, that could affect the accuracy of your answers. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 13 marks)

TOTAL FOR MECHANICS IS 50 MARKS



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Candidate surname

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**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Monday 19 October 2020

Afternoon

Paper Reference **9MA0/32**

Mathematics

Advanced

Paper 32: Mechanics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Information

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- The total mark for this part of the examination is 50. There are 5 questions.
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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

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(Total for Question 1 is 9 marks)



2. A particle P moves with acceleration $(4\mathbf{i} - 5\mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P is moving with velocity $(-2\mathbf{i} + 2\mathbf{j})\text{ms}^{-1}$

(a) Find the velocity of P at time $t = 2$ seconds.

(2)

At time $t = 0$, P passes through the origin O .

At time $t = T$ seconds, where $T > 0$, the particle P passes through the point A .

The position vector of A is $(\lambda\mathbf{i} - 4.5\mathbf{j})\text{m}$ relative to O , where λ is a constant.

(b) Find the value of T .

(4)

(c) Hence find the value of λ

(2)



Question 2 continued

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(Total for Question 2 is 8 marks)



3. (i) At time t seconds, where $t \geq 0$, a particle P moves so that its acceleration $\mathbf{a} \text{ m s}^{-2}$ is given by

$$\mathbf{a} = (1 - 4t)\mathbf{i} + (3 - t^2)\mathbf{j}$$

At the instant when $t = 0$, the velocity of P is $36\mathbf{i} \text{ m s}^{-1}$

- (a) Find the velocity of P when $t = 4$

(3)

- (b) Find the value of t at the instant when P is moving in a direction perpendicular to $\mathbf{i}$

(3)

- (ii) At time t seconds, where $t \geq 0$, a particle Q moves so that its position vector $\mathbf{r}$ metres, relative to a fixed origin O , is given by

$$\mathbf{r} = (t^2 - t)\mathbf{i} + 3t\mathbf{j}$$

Find the value of t at the instant when the speed of Q is 5 m s^{-1}

(6)



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Question 3 continued

Handwriting practice area with 30 horizontal lines.



Question 3 continued

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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 12 marks)



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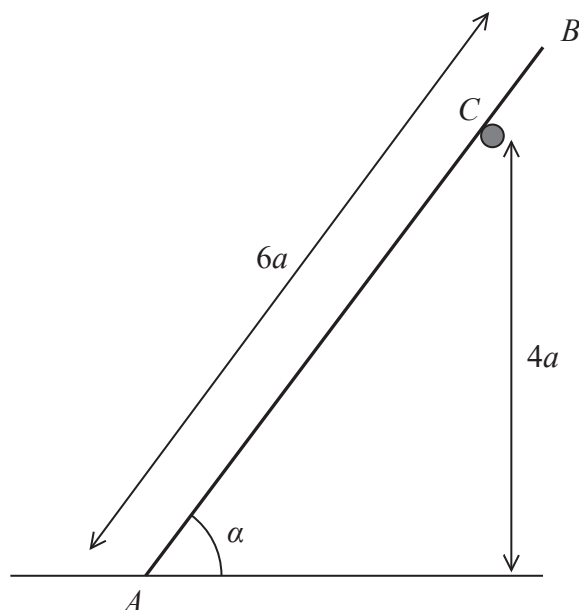


Figure 1

A ladder AB has mass M and length $6a$.

The end A of the ladder is on rough horizontal ground.

The ladder rests against a fixed smooth horizontal rail at the point C .

The point C is at a vertical height $4a$ above the ground.

The vertical plane containing AB is perpendicular to the rail.

The ladder is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{4}{5}$, as shown in Figure 1.

The coefficient of friction between the ladder and the ground is μ .

The ladder rests in limiting equilibrium.

The ladder is modelled as a uniform rod.

Using the model,

(a) show that the magnitude of the force exerted on the ladder by the rail at C is $\frac{9Mg}{25}$ (3)

(b) Hence, or otherwise, find the value of μ . (7)



Question 4 continued

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Question 4 continued

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Question 4 continued

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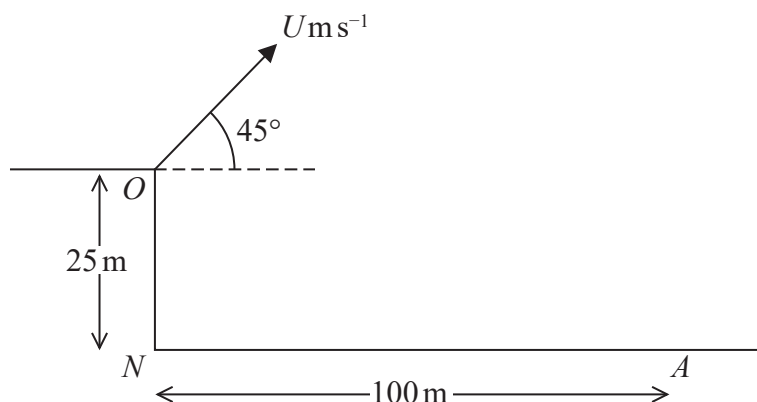
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(Total for Question 4 is 10 marks)



5.

**Figure 2**

A small ball is projected with speed $U \text{ m s}^{-1}$ from a point O at the top of a vertical cliff.

The point O is 25 m vertically above the point N which is on horizontal ground.

The ball is projected at an angle of 45° above the horizontal.

The ball hits the ground at a point A , where $AN = 100 \text{ m}$, as shown in Figure 2.

The motion of the ball is modelled as that of a particle moving freely under gravity.

Using this initial model,

- (a) show that $U = 28$ (6)
- (b) find the greatest height of the ball above the horizontal ground NA . (3)

In a refinement to the model of the motion of the ball from O to A , the effect of air resistance is included.

This refined model is used to find a new value of U .

- (c) How would this new value of U compare with 28, the value given in part (a)? (1)
- (d) State one further refinement to the model that would make the model more realistic. (1)



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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 11 marks)

TOTAL FOR MECHANICS IS 50 MARKS



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Pearson Edexcel Level 3 GCE

Monday 18 October 2021 – Afternoon

Paper
reference

9MA0/32

Mathematics

Advanced

PAPER 32: Mechanics

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1. A particle P moves with constant acceleration $(2\mathbf{i} - 3\mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P is moving with velocity $4\mathbf{i}\text{ms}^{-1}$

- (a) Find the velocity of P at time $t = 2$ seconds.

(2)

At time $t = 0$, the position vector of P relative to a fixed origin O is $(\mathbf{i} + \mathbf{j})\text{m}$.

- (b) Find the position vector of P relative to O at time $t = 3$ seconds.

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



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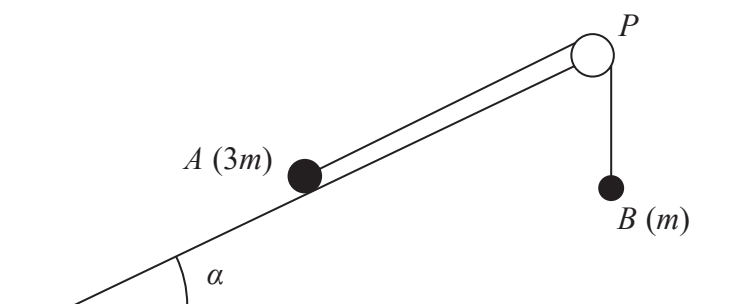


Figure 1

A small stone A of mass $3m$ is attached to one end of a string.

A small stone B of mass m is attached to the other end of the string.

Initially A is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

The string passes over a pulley P that is fixed at the top of the plane.

The part of the string from A to P is parallel to a line of greatest slope of the plane.

Stone B hangs freely below P , as shown in Figure 1.

The coefficient of friction between A and the plane is $\frac{1}{6}$

Stone A is released from rest and begins to move down the plane.

The stones are modelled as particles.

The pulley is modelled as being small and smooth.

The string is modelled as being light and inextensible.

Using the model for the motion of the system before B reaches the pulley,

(a) write down an equation of motion for A (2)

(b) show that the acceleration of A is $\frac{1}{10}g$ (7)

(c) sketch a velocity-time graph for the motion of B , from the instant when A is released from rest to the instant just before B reaches the pulley, explaining your answer. (2)

In reality, the string is not light.

(d) State how this would affect the working in part (b). (1)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 12 marks)



3.

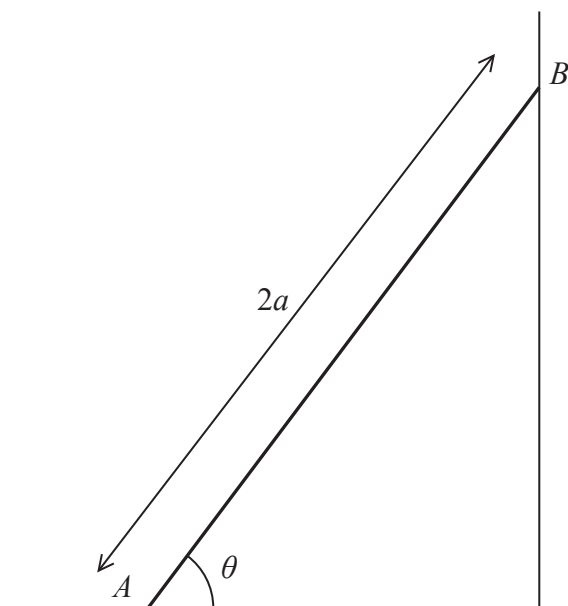


Figure 2

A beam AB has mass m and length $2a$.

The beam rests in equilibrium with A on rough horizontal ground and with B against a smooth vertical wall.

The beam is inclined to the horizontal at an angle θ , as shown in Figure 2.

The coefficient of friction between the beam and the ground is μ

The beam is modelled as a uniform rod resting in a vertical plane that is perpendicular to the wall.

Using the model,

- (a) show that $\mu \geq \frac{1}{2} \cot \theta$ (5)

A horizontal force of magnitude kmg , where k is a constant, is now applied to the beam at A .

This force acts in a direction that is perpendicular to the wall and towards the wall.

Given that $\tan \theta = \frac{5}{4}$, $\mu = \frac{1}{2}$ and the beam is now in limiting equilibrium,

- (b) use the model to find the value of k . (5)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



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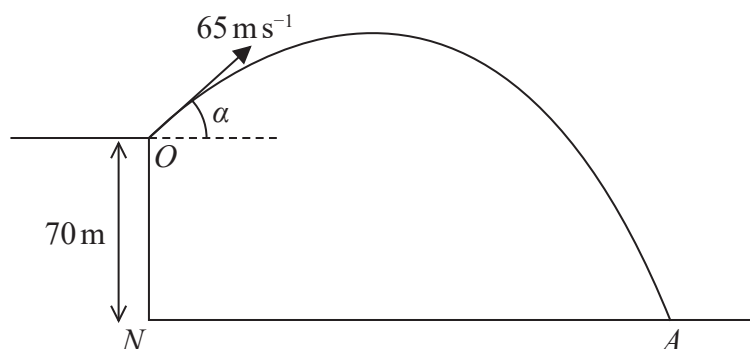


Figure 3

A small stone is projected with speed 65 m s^{-1} from a point O at the top of a vertical cliff.

Point O is 70 m vertically above the point N .

Point N is on horizontal ground.

The stone is projected at an angle α above the horizontal, where $\tan \alpha = \frac{5}{12}$

The stone hits the ground at the point A , as shown in Figure 3.

The stone is modelled as a particle moving freely under gravity.

The acceleration due to gravity is modelled as having magnitude 10 m s^{-2}

Using the model,

(a) find the time taken for the stone to travel from O to A , (4)

(b) find the speed of the stone at the instant just before it hits the ground at A . (5)

One limitation of the model is that it ignores air resistance.

(c) State one other limitation of the model that could affect the reliability of your answers. (1)

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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 10 marks)



5. At time t seconds, a particle P has velocity $\mathbf{v} \text{ m s}^{-1}$, where

$$\mathbf{v} = 3t^{\frac{1}{2}} \mathbf{i} - 2t \mathbf{j} \quad t > 0$$

(a) Find the acceleration of P at time t seconds, where $t > 0$

(b) Find the value of t at the instant when P is moving in the direction of $\mathbf{i} - \mathbf{j}$ (3)

At time t seconds, where $t > 0$, the position vector of P , relative to a fixed origin O , is $\mathbf{r}$ metres.

When $t = 1$, $\mathbf{r} = -\mathbf{j}$

(c) Find an expression for $\mathbf{r}$ in terms of t . (3)

(d) Find the exact distance of P from O at the instant when P is moving with speed 10 m s^{-1}

(6)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

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(Total for Question 5 is 14 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Paper reference	9MA0/32
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Mathematics

Advanced

PAPER 32: Mechanics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Q:1/1/1/1/



1. [In this question, position vectors are given relative to a fixed origin.]

At time t seconds, where $t > 0$, a particle P has velocity $\mathbf{v} \text{ m s}^{-1}$ where

$$\mathbf{v} = 3t^2\mathbf{i} - 6t^{\frac{1}{2}}\mathbf{j}$$

- (a) Find the speed of P at time $t = 2$ seconds. (2)

- (b) Find an expression, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, for the acceleration of P at time t seconds, where $t > 0$ (2)

At time $t = 4$ seconds, the position vector of P is $(\mathbf{i} - 4\mathbf{j}) \text{ m}$.

- (c) Find the position vector of P at time $t = 1$ second. (4)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)



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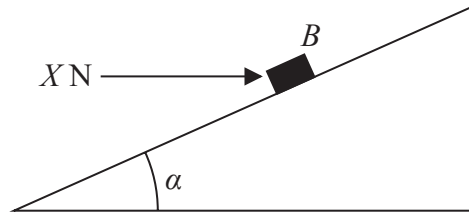


Figure 1

A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A small block B of mass 5 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 1.

The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

The block B is modelled as a particle.

The magnitude of the normal reaction of the plane on B is 68.6 N.

Using the model,

- (a) (i) find the magnitude of the frictional force acting on B ,
- (3)**

- (ii) state the direction of the frictional force acting on B .
- (1)**

The horizontal force of magnitude X newtons is now removed and B moves down the plane.

Given that the coefficient of friction between B and the plane is 0.5

- (b) find the acceleration of B down the plane. (6)



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Question 2 continued

Handwriting practice area with 25 horizontal lines.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 10 marks)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



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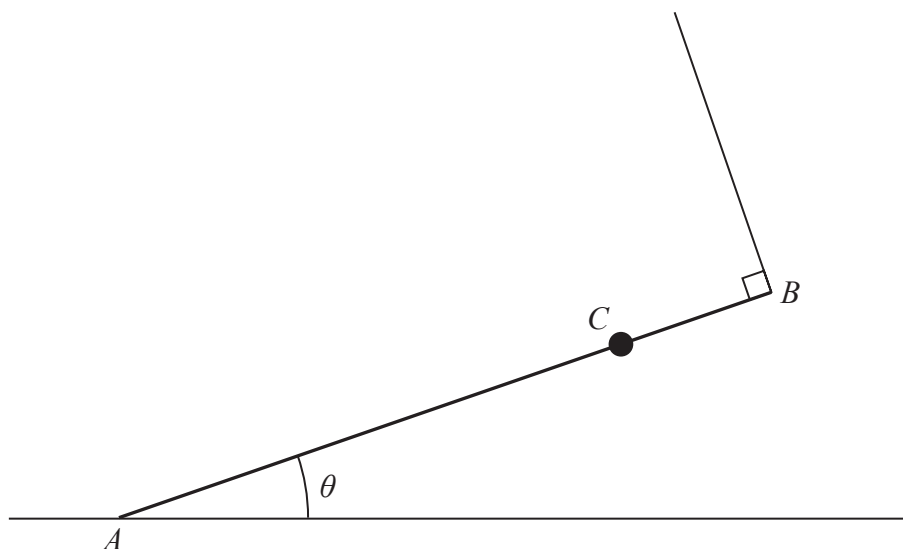


Figure 2

A uniform rod AB has mass M and length $2a$

A particle of mass $2M$ is attached to the rod at the point C , where $AC = 1.5a$

The rod rests with its end A on rough horizontal ground.

The rod is held in equilibrium at an angle θ to the ground by a light string that is attached to the end B of the rod.

The string is perpendicular to the rod, as shown in Figure 2.

- (a) Explain why the frictional force acting on the rod at A acts horizontally to the right on the diagram.

(1)

The tension in the string is T

- (b) Show that $T = 2Mg \cos \theta$

(3)

Given that $\cos \theta = \frac{3}{5}$

- (c) show that the magnitude of the vertical force exerted by the ground on the rod at A is $\frac{57Mg}{25}$

(3)

The coefficient of friction between the rod and the ground is μ

Given that the rod is in limiting equilibrium,

- (d) show that $\mu = \frac{8}{19}$

(4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 11 marks)



5.

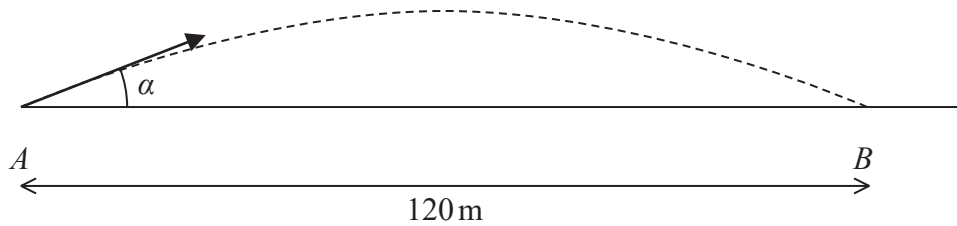


Figure 3

A golf ball is at rest at the point A on horizontal ground.

The ball is hit and initially moves at an angle α to the ground.

The ball first hits the ground at the point B , where $AB = 120\text{ m}$, as shown in Figure 3.

The motion of the ball is modelled as that of a particle, moving freely under gravity, whose initial speed is $U \text{ m s}^{-1}$

Using this model,

- (a) show that $U^2 \sin \alpha \cos \alpha = 588$ (6)

The ball reaches a maximum height of 10m above the ground.

- (b) Show that $U^2 = 1960$ (4)

In a refinement to the model, the effect of air resistance is included.

The motion of the ball, from A to B , is now modelled as that of a particle whose initial speed is $V \text{ m s}^{-1}$

This refined model is used to calculate a value for V

- (c) State which is greater, U or V , giving a reason for your answer. (1)

- (d) State one further refinement to the model that would make the model more realistic. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 12 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Tuesday 20 June 2023

Afternoon

Paper reference **9MA0/32**

Mathematics

Advanced

PAPER 32: Mechanics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 6 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. A car is initially at rest on a straight horizontal road.

The car then accelerates along the road with a constant acceleration of 3.2 m s^{-2}

Find

- (a) the speed of the car after 5 s, (1)

- (b) the distance travelled by the car in the first 5 s. (2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 3 marks)



2.



Figure 1

A particle P has mass 5 kg.

The particle is pulled along a rough horizontal plane by a horizontal force of magnitude 28 N.

The only resistance to motion is a frictional force of magnitude F newtons, as shown in Figure 1.

- (a) Find the magnitude of the normal reaction of the plane on P (1)

The particle is accelerating along the plane at 1.4 m s^{-2}

- (b) Find the value of F (2)

The coefficient of friction between P and the plane is μ

- (c) Find the value of μ , giving your answer to 2 significant figures. (1)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3. At time t seconds, where $t \geq 0$, a particle P has velocity $\mathbf{v} \text{ m s}^{-1}$ where

$$\mathbf{v} = (t^2 - 3t + 7)\mathbf{i} + (2t^2 - 3)\mathbf{j}$$

Find

- (a) the speed of P at time $t = 0$ (3)
- (b) the value of t when P is moving parallel to $(\mathbf{i} + \mathbf{j})$ (2)
- (c) the acceleration of P at time t seconds (2)
- (d) the value of t when the direction of the acceleration of P is perpendicular to $\mathbf{i}$ (2)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors and position vectors are given relative to a fixed origin O]

A particle P is moving on a smooth horizontal plane.

The particle has constant acceleration $(2.4\mathbf{i} + \mathbf{j})\text{ m s}^{-2}$

At time $t = 0$, P passes through the point A .

At time $t = 5$ s, P passes through the point B .

The velocity of P as it passes through A is $(-16\mathbf{i} - 3\mathbf{j})\text{ m s}^{-1}$

- (a) Find the speed of P as it passes through B . (4)

The position vector of A is $(44\mathbf{i} - 10\mathbf{j})\text{ m}$.

At time $t = T$ seconds, where $T > 5$, P passes through the point C .

The position vector of C is $(4\mathbf{i} + c\mathbf{j})\text{ m}$.

- (b) Find the value of T . (3)

- (c) Find the value of c . (3)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 10 marks)



5.

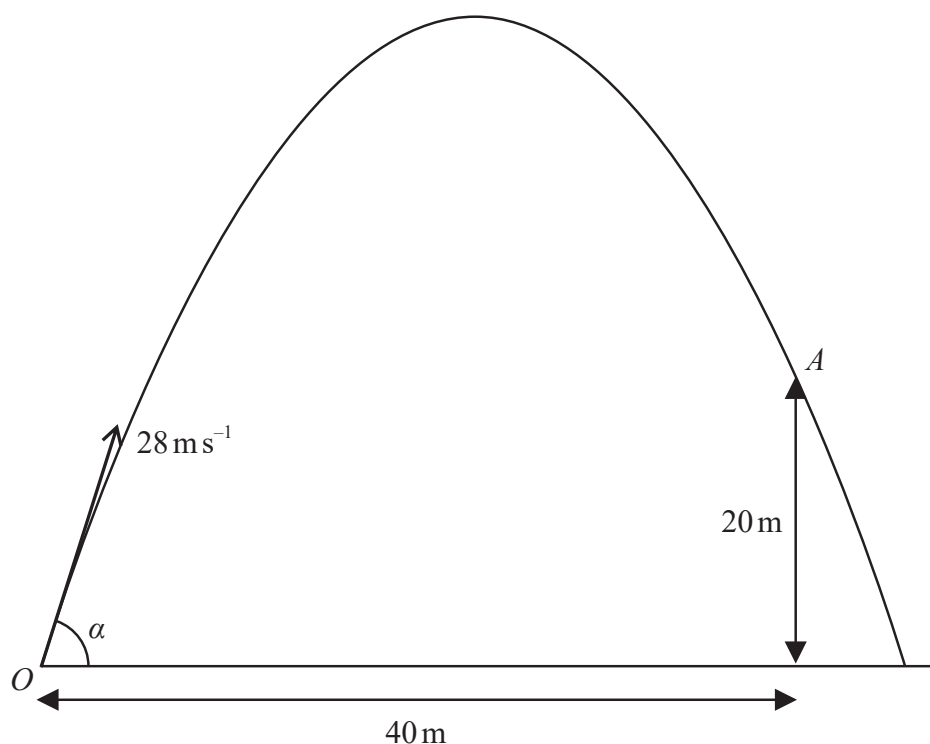


Figure 2

A small ball is projected with speed 28 m s^{-1} from a point O on horizontal ground.

After moving for T seconds, the ball passes through the point A .

The point A is 40 m horizontally and 20 m vertically from the point O , as shown in Figure 2.

The motion of the ball from O to A is modelled as that of a particle moving freely under gravity.

Given that the ball is projected at an angle α to the ground, use the model to

(a) show that $T = \frac{10}{7 \cos \alpha}$ (2)

(b) show that $\tan^2 \alpha - 4 \tan \alpha + 3 = 0$ (5)

(c) find the greatest possible height, in metres, of the ball above the ground as the ball moves from O to A . (3)

The model does not include air resistance.

(d) State one other limitation of the model. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 11 marks)



6.

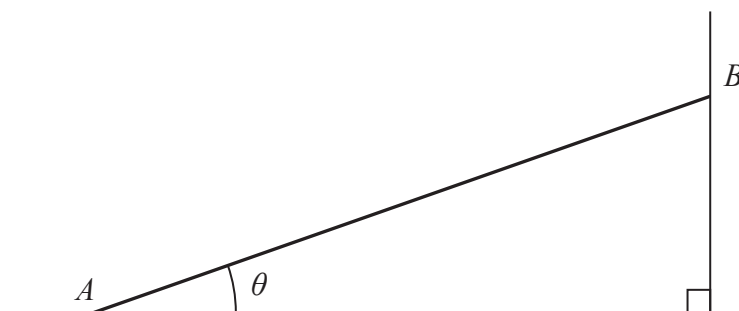


Figure 3

A rod AB has mass M and length $2a$.

The rod has its end A on rough horizontal ground and its end B against a smooth vertical wall.

The rod makes an angle θ with the ground, as shown in Figure 3.

The rod is at rest in limiting equilibrium.

- (a) State the direction (left or right on Figure 3 above) of the frictional force acting on the rod at A . **Give a reason for your answer.**

(1)

The magnitude of the normal reaction of the wall on the rod at B is S .

In an initial model, the rod is modelled as being **uniform**.

Use this initial model to answer parts (b), (c) and (d).

- (b) By taking moments about A , show that

$$S = \frac{1}{2} Mg \cot \theta$$

(3)

The coefficient of friction between the rod and the ground is μ

Given that $\tan \theta = \frac{3}{4}$

- (c) find the value of μ

(5)

- (d) find, in terms of M and g , the magnitude of the resultant force acting on the rod at A .

(3)

In a new model, the rod is modelled as being **non-uniform**, with its centre of mass closer to B than it is to A .

A new value for S is calculated using this new model, with $\tan \theta = \frac{3}{4}$

- (e) State whether this new value for S is larger, smaller or equal to the value that S would take using the initial model. **Give a reason for your answer.**

(1)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 13 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Thursday 20 June 2024

Afternoon	Paper reference	9MA0/32
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Mathematics

Advanced

PAPER 32: Mechanics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 6 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

**Figure 1**

Figure 1 shows a particle P of mass 0.5 kg at rest on a rough horizontal plane.

(a) Find the magnitude of the normal reaction of the plane on P .

(1)

The coefficient of friction between P and the plane is $\frac{2}{7}$

A horizontal force of magnitude X newtons is applied to P .

Given that P is now in limiting equilibrium,

(b) find the value of X .

(2)

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 3 marks)



2.

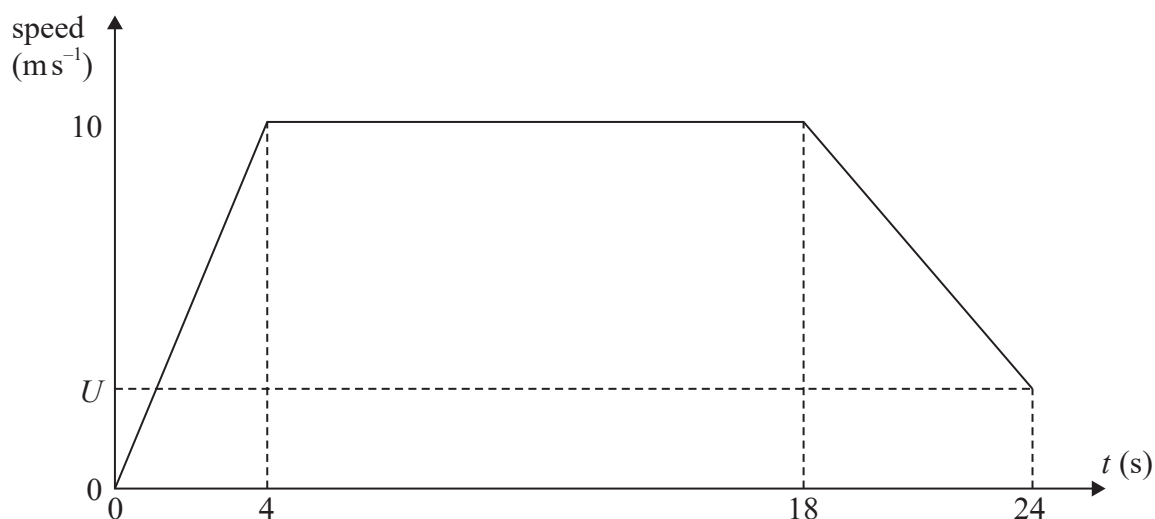


Figure 2

Figure 2 shows a speed-time graph for a model of the motion of an athlete running a **200 m** race in 24 s.

The athlete

- starts from rest at time $t = 0$ and accelerates at a constant rate, reaching a speed of 10 ms^{-1} at $t = 4$
- then moves at a constant speed of 10 ms^{-1} from $t = 4$ to $t = 18$
- then decelerates at a constant rate from $t = 18$ to $t = 24$, crossing the finishing line with speed $U \text{ ms}^{-1}$

Using the model,

- (a) find the acceleration of the athlete during the first 4 s of the race, stating the units of your answer,

(2)

- (b) find the distance covered by the athlete during the first 18 s of the race,

(3)

- (c) find the value of U .

(3)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3.

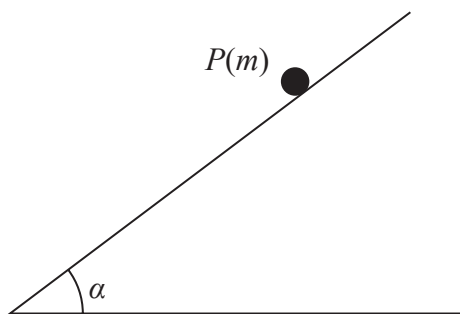


Figure 3

A particle P of mass m is held at rest at a point on a rough inclined plane, as shown in Figure 3.

It is given that

- the plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{5}{12}$
- the coefficient of friction between P and the plane is μ , where $\mu < \frac{5}{12}$

The particle P is released from rest and slides down the plane.
Air resistance is modelled as being negligible.

Using the model,

- (a) find, in terms of m and g , the magnitude of the normal reaction of the plane on P , (2)

- (b) show that, as P slides down the plane, the acceleration of P down the plane is

$$\frac{1}{13}g(5 - 12\mu) \quad (4)$$

- (c) State what would happen to P if it is released from rest but $\mu \geq \frac{5}{12}$ (1)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 7 marks)



4.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

[In this question, $\mathbf{i}$ is a unit vector due east and $\mathbf{j}$ is a unit vector due north.
Position vectors are given relative to a fixed origin O .]

At time t seconds, $t \geq 1$, the position vector of a particle P is $\mathbf{r}$ metres, where

$$\mathbf{r} = ct^{\frac{1}{2}}\mathbf{i} - \frac{3}{8}t^2\mathbf{j}$$

and c is a constant.

When $t = 4$, the bearing of P from O is 135°

- (a) Show that $c = 3$ (3)

- (b) Find the speed of P when $t = 4$ (4)

When $t = T$, P is accelerating in the direction of $(-\mathbf{i} - 27\mathbf{j})$.

- (c) Find the value of T . (4)



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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 11 marks)



5.

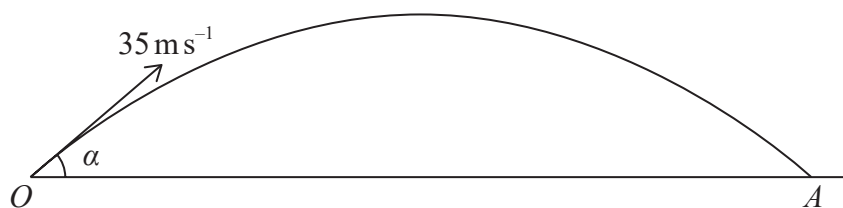


Figure 4

At time $t = 0$, a small stone is projected with velocity 35 ms^{-1} from a point O on horizontal ground.

The stone is projected at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

In an initial model

- the stone is modelled as a particle P moving freely under gravity
- the stone hits the ground at the point A

Figure 4 shows the path of P from O to A .

For the motion of P from O to A

- at time t seconds, the horizontal distance of P from O is x metres
- at time t seconds, the vertical distance of P above the ground is y metres

(a) Using the model, show that

$$y = \frac{3}{4}x - \frac{1}{160}x^2 \quad (6)$$

(b) Use the answer to (a), or otherwise, to find the length OA . (2)

Using the model, the greatest height of the stone above the ground is found to be H metres.

(c) Use the answer to (a), or otherwise, to find the value of H . (2)

- The model is refined to include air resistance.

Using this new model, the greatest height of the stone above the ground is found to be K metres.

(d) State which is greater, H or K , justifying your answer. (1)

(e) State one limitation of this refined model. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 12 marks)



6.

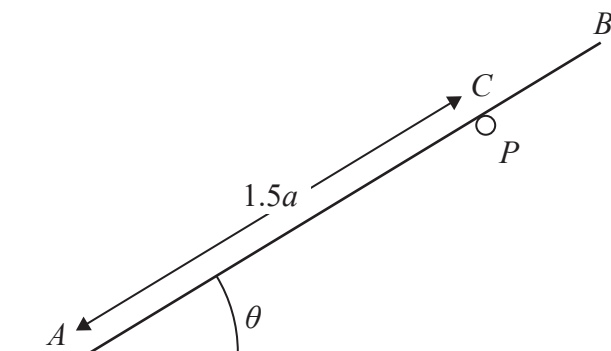


Figure 5

Figure 5 shows a uniform rod AB of mass M and length $2a$.

- the rod has its end A on rough horizontal ground
- the rod rests in equilibrium against a small smooth fixed horizontal peg P
- the point C on the rod, where $AC = 1.5a$, is the point of contact between the rod and the peg
- the rod is at an angle θ to the ground, where $\tan \theta = \frac{4}{3}$

The rod lies in a vertical plane perpendicular to the peg.

The magnitude of the normal reaction of the peg on the rod at C is S .

- (a) Show that $S = \frac{2}{5}Mg$ (3)

The coefficient of friction between the rod and the ground is μ .

Given that the rod is in limiting equilibrium,

- (b) find the value of μ . (6)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 9 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

Pearson Edexcel Level 3 GCE

Thursday 19 June 2025

Afternoon

Paper reference

9MA0/32

Mathematics

Advanced

PAPER 32: Mechanics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 6 questions.
- The marks for **each** question are shown in brackets
 – use this as a guide as to how much time to spend on each question.

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. A car moves in a straight line along a horizontal road with constant acceleration 2 m s^{-2}

The car is moving with speed 15 m s^{-1} in the direction of the acceleration when it passes a signpost on the road.

The car is modelled as a particle.

- (a) Use the model to find the speed of the car 4 s after passing the signpost.

(2)

Figure 1 below shows the horizontal forces acting on the car.

Given that

- the car has mass 800 kg
- the driving force of the engine has magnitude D newtons
- the resistance to the motion of the car has magnitude 400 N
- the acceleration of the car is 2 m s^{-2} in the direction of the driving force

- (b) use the model to find the value of D .

(2)

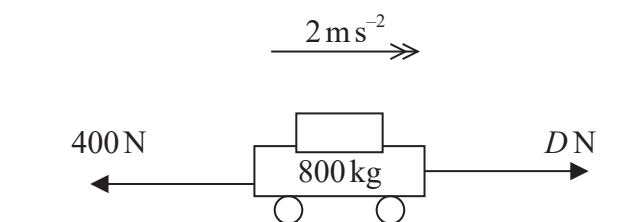


Figure 1

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2.



Figure 2

A small box B of mass 2 kg is dragged in a straight line, along a **rough** horizontal plane, at a **constant speed** by a force of magnitude 5 N .

The line of action of the force makes an angle α with the plane, where $\sin \alpha = \frac{3}{5}$, as shown in Figure 2.

- (a) Show that the magnitude of the normal reaction of the plane on the box is 16.6 N . (3)

At the instant when B is at the point O on the plane, the force of magnitude 5 N is removed.

- (b) Describe the motion of the box after the force of magnitude 5 N is removed. (1)
- (c) Find the magnitude of the normal reaction of the plane on the box after the force of magnitude 5 N is removed. (1)

Given that after the force of magnitude 5 N is removed

- the box is modelled as a particle
- air resistance is modelled as being negligible
- the coefficient of friction between the box and the plane is modelled as 0.2
- the speed of the box as it passes through O is 4 m s^{-1}
- the box comes to rest at the point X on the plane

- (d) use the model to find the length OX . (4)
- (e) State one limitation of the model, apart from ignoring air resistance, that could affect your answer to part (d). (1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 10 marks)



3. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors due east and due north respectively.]

A particle P of mass 0.5 kg moves with constant acceleration $(2\mathbf{i} - 2.4\mathbf{j}) \text{ m s}^{-2}$ on a smooth horizontal plane under the action of a constant horizontal force $\mathbf{F} \text{ N}$.

- (a) Find $\mathbf{F}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

(1)

At time $t = 0$, P is moving with velocity $(-7\mathbf{i} + 7.8\mathbf{j}) \text{ m s}^{-1}$

- (b) Find the velocity of P at time $t = 2$ seconds.

(2)

- (c) Find the direction of motion of P at time $t = 2$ seconds, giving your answer as a bearing in degrees.

(3)

At time $t = 0$, P passes through the point O .

At time $t = 5$ seconds, P passes through the point A .

- (d) Find $\overrightarrow{OA}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

(2)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 8 marks)



4.

At time t seconds, where $t > 0$, the position vector of a particle P is $\mathbf{r}$ metres where

$$\mathbf{r} = 4t^{\frac{3}{2}}\mathbf{i} - t^2\mathbf{j}$$

- (a) Find the position vector of P at $t = 4$ (1)

- (b) Find the exact distance of P from O at $t = 4$

- (c) Find an expression for the velocity of P at time t seconds, where $t > 0$, giving your answer in terms of t , $\mathbf{i}$ and $\mathbf{j}$
- (2)

At $t = T$, the acceleration of P is in a direction that is perpendicular to the line with equation $y = \frac{1}{3}x$

- (d) Find the value of T . (4)

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



5.

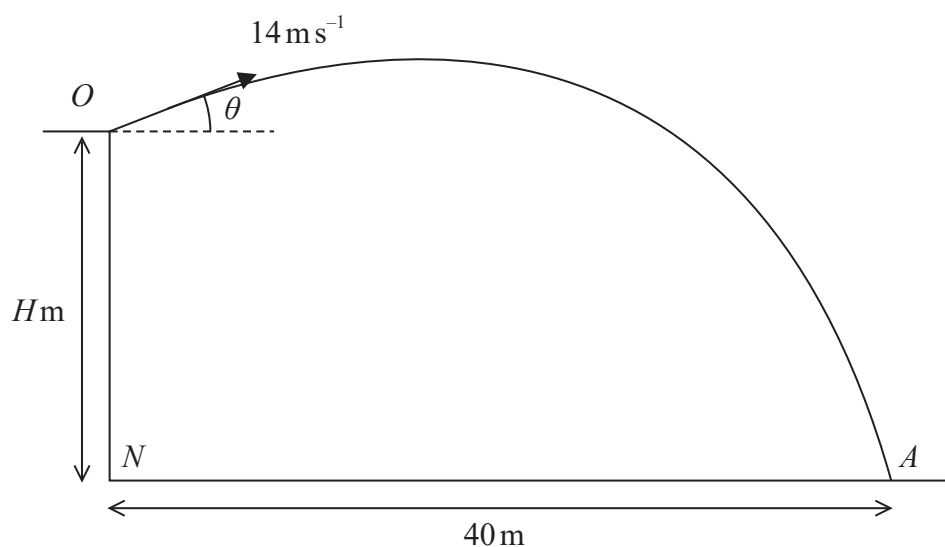


Figure 3

A small stone is projected with speed 14 ms^{-1} from a point O on the top of a cliff. The point O is H metres vertically above the point N .

Point N is on horizontal ground.

The stone is projected at an angle θ above the horizontal, where $\tan \theta = \frac{1}{2}$

The stone strikes the horizontal ground at the point A , where $NA = 40\text{ m}$, as shown in Figure 3.

The stone is modelled as a particle moving freely under gravity.

Using this model, find

(a) the value of H (5)

(b) the maximum height of the stone above the horizontal ground. (3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



6.

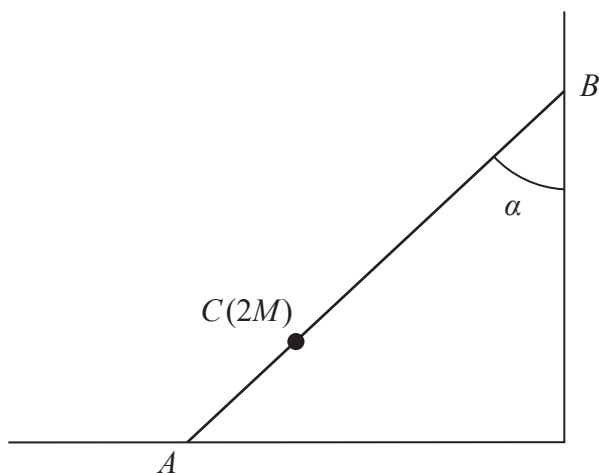


Figure 4

A uniform rod AB has mass M and length $2a$.

A particle of mass $2M$ is attached to the rod at the point C , where $AC = 0.5a$

The rod rests with end A on rough horizontal ground and end B against a vertical wall.

The rod lies in a vertical plane which is perpendicular to the wall.

The rod is in equilibrium at an angle α to the wall, as shown in Figure 4.

In an initial model

- the vertical wall is modelled as being **smooth**
- the magnitude of the normal reaction of the ground on the rod at A is R
- the magnitude of the force exerted on the rod by the wall at B is S

Using the model,

(a) find R in terms of M and g (1)

(b) show that $S = Mg \tan \alpha$ (3)

In a refined model

- the vertical wall is modelled as being **rough**
 - the magnitude of the normal reaction of the ground on the rod at A is R_1
- (c) State which is greater, R or R_1 , giving a reason for your answer. (1)



Question 6 continued

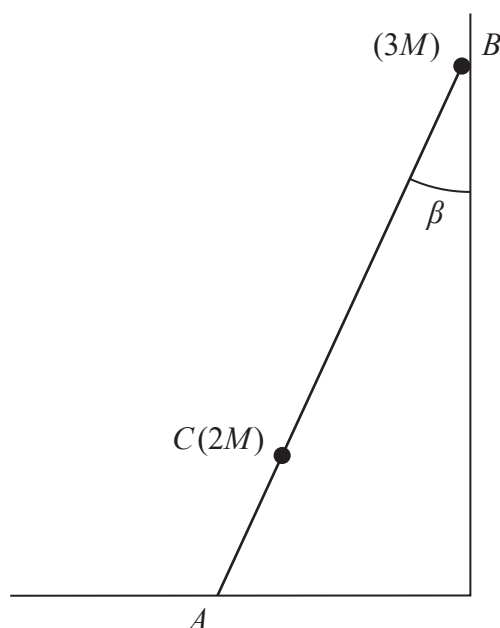


Figure 5

A second particle of mass $3M$ is now attached to the rod at B .

The rod again rests with end A on rough horizontal ground and end B against the vertical wall.

The rod lies in a vertical plane which is perpendicular to the wall.

The rod is now in limiting equilibrium at an angle β to the wall, as shown in Figure 5.

The vertical wall is again modelled as being **smooth**.

The coefficient of friction between the rod and the ground is μ

Given that $\tan \beta = \frac{1}{2}$

(d) use the model to show that $\mu = \frac{1}{3}$

(6)



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 11 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper Set 1			
		Paper Reference 9MA0-32	
Mathematics Advanced Paper 32: Mechanics			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- Unless otherwise stated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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- 1.** A car moves along a straight horizontal road.

The car starts from rest at a fixed point A on the road and moves with constant acceleration for 30 seconds, reaching a speed of 15 m s^{-1} .

This speed is then maintained.

When the car has been moving for 15 seconds a motorbike starts from rest at A and moves along the same road in the same direction as the car.

The motorbike accelerates at 1.5 m s^{-2} so that it catches up with the car when the car has been moving for T seconds.

- (a) Using the same axes, sketch the speed-time graph of the car and the speed-time graph of the motorbike up to the time when the motorbike catches up with the car. (3)
- (b) Find the speed of the motorbike at the instant it catches up with the car. (6)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 9 marks)



2.

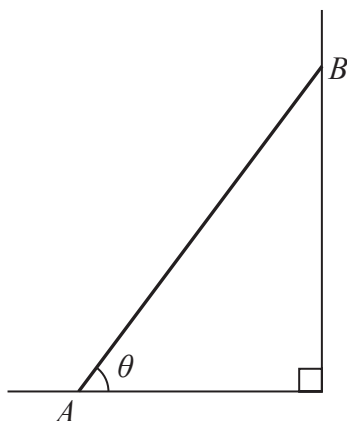


Figure 1

The ladder AB shown in Figure 1 has length $2a$ and weight W .

The ladder rests in equilibrium with end A on rough horizontal ground and end B against a smooth vertical wall.

The ladder rests in a vertical plane perpendicular to the wall, and is inclined at angle θ to the ground.

The coefficient of friction between the ladder and the ground is μ .

The ladder is on the point of slipping.

The ladder is modelled as a uniform rod.

(a) Show that $\mu = \frac{1}{2 \tan \theta}$ (7)

(b) If the ladder were not modelled as uniform, state how this would affect the calculated value of μ , explaining your answer carefully. (2)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 9 marks)



3. [In this question position vectors are given relative to a fixed origin O]

A particle P moves under the action of a single force $\mathbf{F}$ newtons.

At time t seconds, where $t \geq 0$, the position vector of P , $\mathbf{r}$ metres, is given by

$$\mathbf{r} = (t^3 - 5t)\mathbf{i} + (5t^2 + 6t)\mathbf{j}$$

The mass of P is 0.5 kg.

At time T seconds, P is moving in the direction of the vector $(\mathbf{i} + 2\mathbf{j})$.

(a) Find the value of T .

(5)

(b) Find the magnitude of $\mathbf{F}$ when $t = 2$

(4)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4. [In this question the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane, $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upward.]

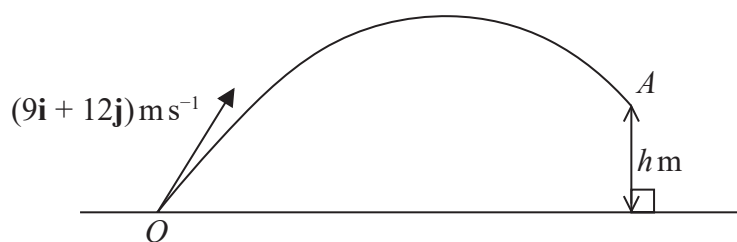


Figure 2

A small ball is projected from the fixed point O on horizontal ground with velocity $(9\mathbf{i} + 12\mathbf{j})\text{ m s}^{-1}$

The ball passes through the point A which is h metres vertically above the level of O , as shown in Figure 2.

The velocity of the ball at the instant it passes through the point A is $\lambda(\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$, where λ is a positive constant.

The ball is modelled as a particle moving freely under gravity.

- Find the value of h . (4)
- State the minimum speed of the ball as it moves from O to A . (1)
- Find the length of time for which the speed of the ball is less than 12 m s^{-1} (4)

The model could be refined by considering air resistance.

- Suggest one other refinement to the model that would make it more realistic. (1)



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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 10 marks)



5.

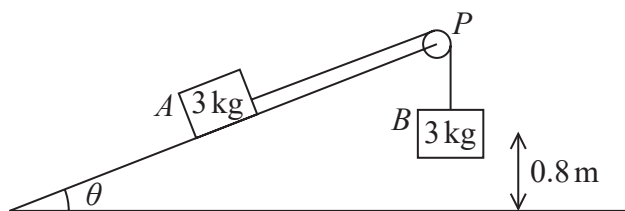


Figure 3

Two packages A and B , each of mass 3 kg , are attached to the ends of a rope.

Initially A is held at rest on a smooth fixed plane that is inclined at angle θ to the horizontal ground, where $\sin \theta = \frac{2}{7}$

The rope passes over a pulley, P , fixed at the top of the plane.

The pulley is modelled as small and smooth.

The part of the string from A to P is parallel to a line of greatest slope of the plane.

Package B hangs freely below P , as shown in Figure 3.

The packages are released from rest with the string taut and A moves up the plane.

In this model, the packages are modelled as particles and the rope as a light inextensible string.

The magnitude of the tension in the string immediately after the packages are released is T newtons.

(a) Find the value of T .

(6)

At the instant when the packages are released from rest, B is 0.8 m above the ground and A is at the point C on the plane.

When B reaches the ground, B is immediately brought to rest by the impact with the ground.

In the subsequent motion, A does not reach P and comes to instantaneous rest at the point D on the plane.

(b) Find the distance CD .

(5)

(c) State two limitations of the model that could affect the reliability of your answers.

(2)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 13 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

--	--	--	--	--

Mock Paper – Set 2

Paper Reference **9MA0/32**

Mathematics

Advanced

Paper 32: Mechanics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. At time t seconds ($t \geq 0$), a particle P is modelled as having velocity $\mathbf{v} \text{ ms}^{-1}$, where

$$\mathbf{v} = (3t^2 - 12t)\mathbf{i} + (9t^2 - 3t)\mathbf{j}$$

and having acceleration $\mathbf{a} \text{ ms}^{-2}$

- (a) Find $\mathbf{a}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

When $t = 0$, P is at the origin O .

At time t seconds ($t \geq 0$), P has position vector $\mathbf{r}$ metres relative to O .

- (b) Find $\mathbf{r}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

At the instant when $\mathbf{a} = \lambda\mathbf{j}$, where λ is a constant, P is at the point A .

- (c) Find the position vector of A relative to O .

(4)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)



2.

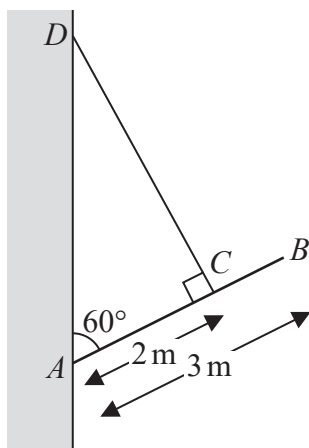


Figure 1

A beam AB has weight 40 N and length 3 m .

The beam is freely hinged at the end A to a vertical wall.

The beam is held in equilibrium at an angle of 60° to the wall by a rope.

One end of the rope is attached to the point C on the beam, where $AC = 2\text{ m}$.

The other end of the rope is attached to a point D on the wall, where D is vertically above A .

The rope is perpendicular to the beam, as shown in Figure 1.

The rope and the beam lie in a vertical plane that is perpendicular to the wall.

The beam is modelled as a uniform rod and the rope as a light inextensible string.

Using the model, find

(a) the tension in the rope, (3)

(b) the magnitude of the resultant force acting on the beam at A . (6)

If the rope was not modelled as being light,

(c) state how this would affect the tension along the rope, explaining your answer. (2)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 11 marks)



3.

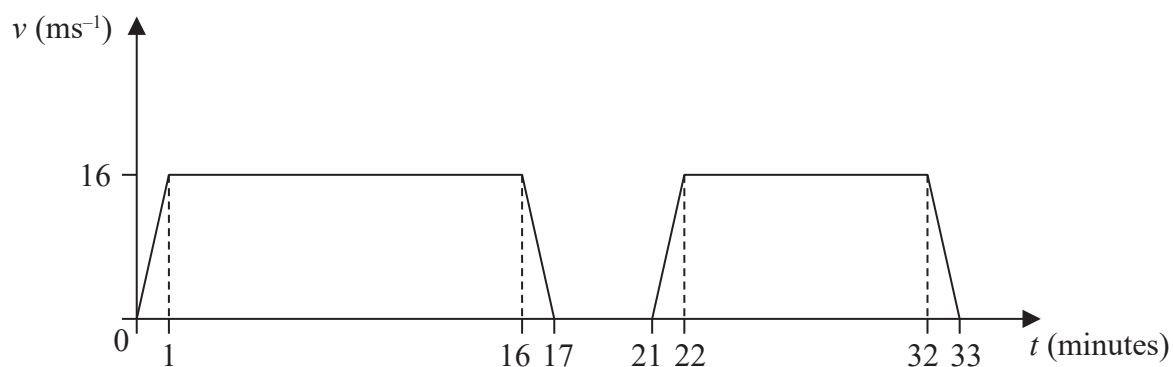


Figure 2

A train X runs on a straight horizontal track that connects stations A and C .

Station B lies between A and C .

Figure 2 shows the graph of the speed, $v \text{ ms}^{-1}$, of train X against the time, t minutes, after noon.

Train X leaves A at noon and accelerates uniformly from rest until $t = 1$, when it is moving with speed 16 ms^{-1} .

Train X then continues to move along the track at constant speed for 15 minutes, before decelerating uniformly and coming to rest at B at time $t = 17$.

Train X leaves B at time $t = 21$ and accelerates uniformly for one minute, reaching a speed of 16 ms^{-1} .

Train X then moves along the track at a constant speed of 16 ms^{-1} for 10 minutes, before decelerating uniformly and coming to rest at C at time $t = 33$.

(a) Find the distance of C from A , stating the units of your answer.

(3)

A second train, Y , leaves A at T minutes after noon and moves in the same direction as train X on a parallel straight horizontal track.

Train Y accelerates uniformly from rest for 2 minutes, reaching a speed of 24 ms^{-1} .

Train Y then moves along the track at a constant speed of 24 ms^{-1} and passes C at this speed.

Train Y passes C at the instant train X stops at C .

(b) Find the value of T .

(5)

(c) State one assumption made in your working that could affect the accuracy of your answer to part (b).

(1)



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Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



4.

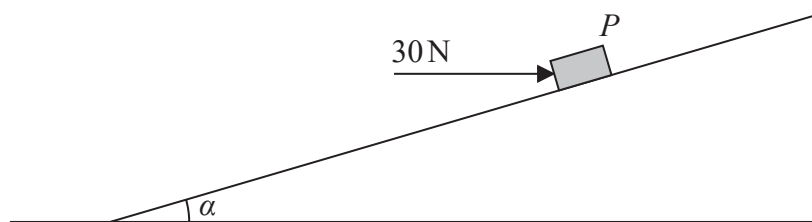


Figure 3

A package P of weight 20 N is moving up an inclined plane under the action of a horizontal force of magnitude 30 N , as shown in Figure 3.

The force is acting in a vertical plane through a line of greatest slope of the plane.

The coefficient of friction between P and the plane is μ

The plane is inclined at angle α to the horizontal, where $\tan \alpha = \frac{5}{12}$

Package P is modelled as a particle.

- (a) Find the magnitude of the normal reaction of the plane on P (2)
- (b) Find the range of possible values of μ (4)

The horizontal force is now removed and P continues to slide up the plane until P comes to instantaneous rest.

Package P then slides back down the plane.

Given that $\mu = \frac{1}{3}$

- (c) find the acceleration of P as it slides down the plane. (5)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 11 marks)



5. A small ball is projected with speed u from a point O on horizontal ground. The angle of projection is θ to the horizontal, where $0 < \theta < 90^\circ$. The ball hits the ground at the point A .

The ball is modelled as a particle moving freely under gravity.

(a) Show that, according to the model, $OA = \frac{u^2 \sin 2\theta}{g}$ (5)

A golfer hits a golf ball with speed 25 ms^{-1} from a point X on horizontal ground.

The golf ball hits the ground at the point Y .

The angle of projection is θ to the horizontal, where $0 < \theta < 90^\circ$

The golfer requires the distance XY to be at least 40 m.

The golf ball is modelled as a particle moving freely under gravity.

- (b) Find, according to the model, the size of the largest possible angle θ (2)

Given that $\theta = 30^\circ$ and that the golf ball is more than 3 m above the ground for T seconds,

- (c) find the value of T . (4)



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Question 5 continued

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DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 11 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mock Paper Set 3

Paper Reference **9MA0/32**

Mathematics
Advanced
Paper 32: Mechanics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 10 marks)



2.

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 0.5 kg is moving on a smooth horizontal plane.

The origin O is on the plane.

At time $t = 0$, P passes through O moving with velocity $(\mathbf{i} - \mathbf{j}) \text{ ms}^{-1}$

At time t seconds, the resultant horizontal force acting on P is

$$[(3t - 1)\mathbf{i} + 2\mathbf{j}] \text{ N}$$

(a) Find the velocity of P at $t = 2$

(5)

(b) Find the distance of P from O at $t = 2$

(4)



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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 9 marks)



3.

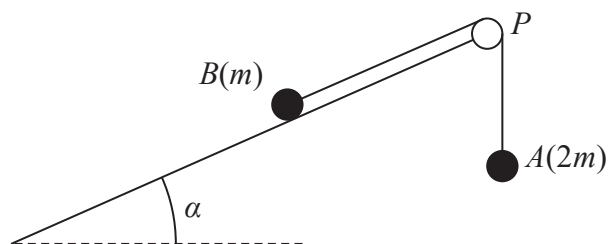


Figure 1

A fixed rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{4}{3}$

A small smooth light pulley P is fixed at the top of the plane.

A particle A of mass $2m$ and a particle B of mass m are connected by a rope.

The rope passes over the pulley P

The part of the rope from B to P is parallel to a line of greatest slope of the plane.

Particle B is held at rest on the rough plane and A hangs freely with the rope taut, as shown in Figure 1.

Particle B is released from rest and begins to move up the plane.

- (a) Find, in terms of mg , the magnitude of the normal reaction that the plane exerts on B .

(2)

The rope is modelled as being light and inextensible.

As B moves up the plane, the tension in the rope has magnitude $\frac{7mg}{5}$

The coefficient of friction between B and the plane is μ .

Using the model,

- (b) (i) write down an equation of motion for A ,
 (ii) hence, find in terms of g , the magnitude of the acceleration of A ,

(3)

- (c) find the value of μ .

(5)

In reality, the rope would not be light.

- (d) State how this would affect the tension in the rope.

(1)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 11 marks)



4. [In this question the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed horizontally and vertically upwards respectively.]

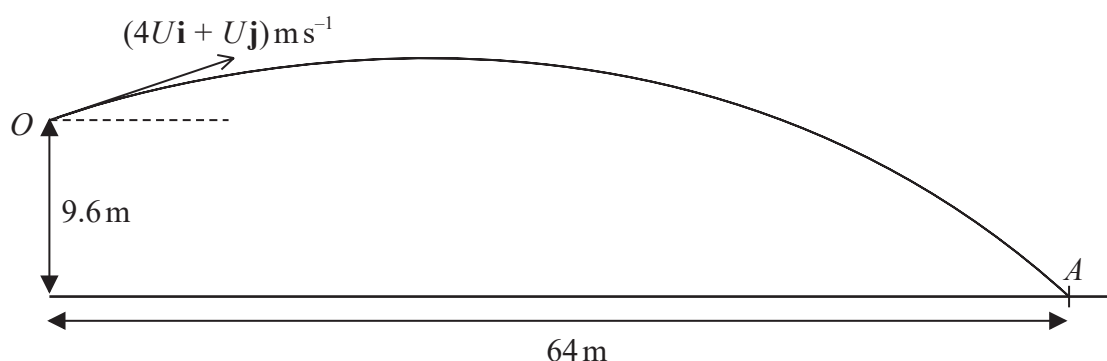


Figure 2

The point O is 9.6 m above horizontal ground.

A small ball is projected with velocity $(4U\mathbf{i} + U\mathbf{j})\text{ m s}^{-1}$, where U is a positive constant, from the point O

The ball first hits the ground T seconds later, at the point A

The point A is at a horizontal distance of 64 m from O , as shown in Figure 2.

In an initial model

- the ball is modelled as a particle moving under gravity
- air resistance is ignored
- the ball has an initial speed of $V\text{ m s}^{-1}$

Using this model,

- show that $UT = 16$ (2)
- find the value of V (6)
- State two improvements to the model, other than including air resistance, that would make the model more realistic. (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 10 marks)



5.

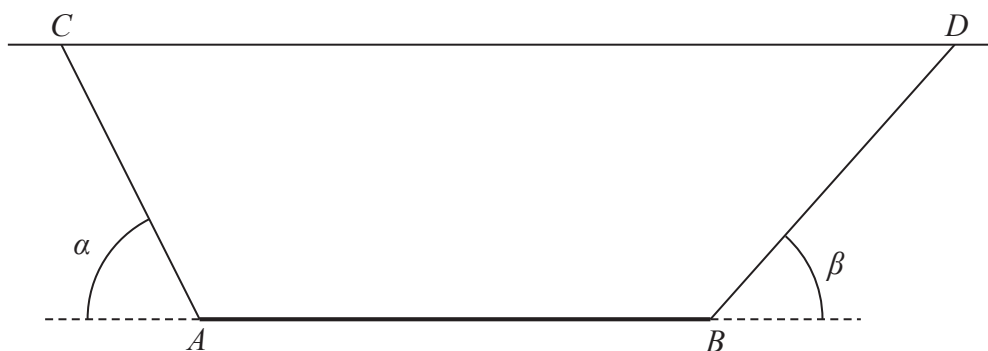


Figure 3

A non-uniform rod AB is held in equilibrium in a horizontal position by two light inextensible strings.

The first string has one end attached to the end A of the rod and the other end attached to a point C on a horizontal ceiling.

The second string has one end attached to the end B of the rod and the other end attached to a point D on the horizontal ceiling, as shown in Figure 3.

The points A , B , C and D all lie in the same vertical plane.

Given that

- the rod AB has weight W and length a
- the centre of mass of the rod is a distance $\frac{1}{3}a$ from A
- the string AC makes an angle α with the horizontal, where $\tan \alpha = \frac{4}{3}$
- the string BD makes an angle β with the horizontal

(a) show that the tension in the string AC is $\frac{5}{6}W$ (3)

The tension in the string BD is S

(b) Show that $S \sin \beta = \frac{1}{3}W$ (2)

(c) Find S in terms of W (5)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 10 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Please check the examination details below before entering your candidate information

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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Paper
reference

9MA0/32

Mathematics

Advanced

PAPER 32: Mechanics

Mock Set 4

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. [In this question, position vectors are given relative to a fixed origin.]

A particle P is moving with constant acceleration $(3\mathbf{i} - 2\mathbf{j})\text{m s}^{-2}$

At time $t = 1\text{ s}$

- P is at the point with position vector $(5\mathbf{i} - 2\mathbf{j})\text{m}$
- P is moving with velocity $(-\mathbf{i} + 4\mathbf{j})\text{m s}^{-1}$

Find

(a) the exact speed of P at time $t = 4\text{ s}$ (4)

(b) the position vector of P at time $t = 3\text{ s}$ (3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. In this question, solutions relying on calculator technology are not acceptable.

[In this question, position vectors are given relative to a fixed origin O .]

A particle P is moving in the xy -plane.

At time t seconds, where $t \geq 0$, particle P is moving with acceleration $\mathbf{a} \text{ m s}^{-2}$ where

$$\mathbf{a} = (1 - 3t)\mathbf{i} + (2t^2 - 2t)\mathbf{j}$$

At time $t = 0$

- P passes through the point with position vector $(\mathbf{i} + \mathbf{j})\text{m}$
- P is moving with velocity $(3\mathbf{i} - 2\mathbf{j})\text{m s}^{-1}$

(a) Find the velocity of P at time t seconds, where $t \geq 0$

(3)

(b) Find the position vector of P relative to O , when the acceleration of P is parallel to $(-\mathbf{i} - \mathbf{j})$.

(6)



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Question 2 continued

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(Total for Question 2 is 9 marks)



3.

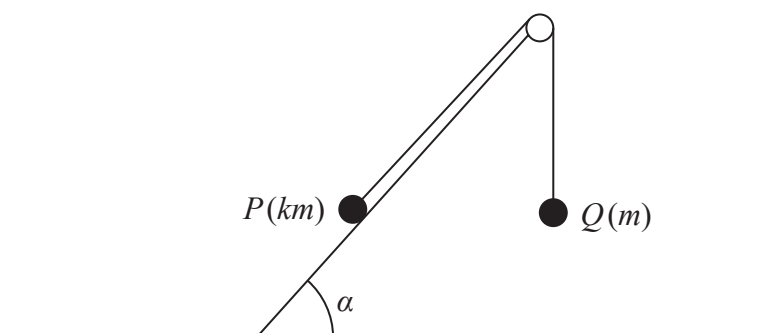


Figure 1

One end of a light inextensible string is attached to a particle P of mass km , where $k > 1.25$

The other end of the string is attached to a particle Q of mass m .

The string passes over a small smooth light pulley that is fixed at the top of a plane.

The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{4}{3}$

Particle P is held at rest on the plane and particle Q hangs at rest with the string taut, as shown in Figure 1.

The part of the string from P to the pulley lies along a line of greatest slope of the plane.

The two particles and the pulley all lie in the same vertical plane.

The particle P is released from rest.

In an initial model,

- the plane is modelled as being smooth
- P slides down the plane with acceleration $\frac{1}{5}g$

Using this model,

(a) write down an equation of motion for P

(2)

(b) find the value of k .

(4)

In a second model,

- the plane is modelled as being rough
- the coefficient of friction between P and the plane is μ
- P remains at rest but is on the point of slipping down the plane

Using this model,

(c) find, in terms of k , m and g , the magnitude of the normal reaction exerted by the plane on P .

(2)

(d) find, in terms of k , the value of μ .

(6)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 14 marks)



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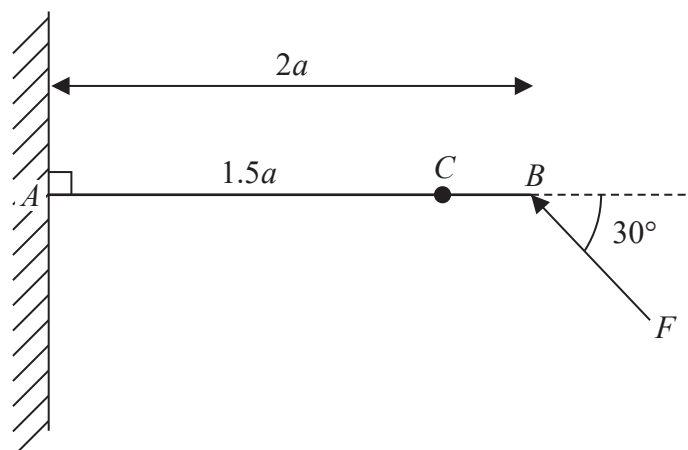


Figure 2

A uniform rod AB has mass m and length $2a$.

A particle of mass $2m$ is attached to the rod at the point C , where $AC = 1.5a$

The rod is freely hinged at its end A to a fixed vertical wall.

The rod is held in equilibrium in a horizontal position by a force applied to its end B .

The force has magnitude F and acts at 30° to the horizontal, as shown in Figure 2.

The line of action of the force and the rod lie in the same vertical plane that is perpendicular to the wall.

(a) Show that $F = 4mg$ (3)

(b) Find the magnitude, in terms of m and g , of the vertical component of the force exerted on the rod by the wall at A . (2)

The line of action of the force exerted on the rod by the wall at A makes an angle θ with the horizontal.

(c) Find the exact value of $\tan \theta$. (4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 9 marks)



5.

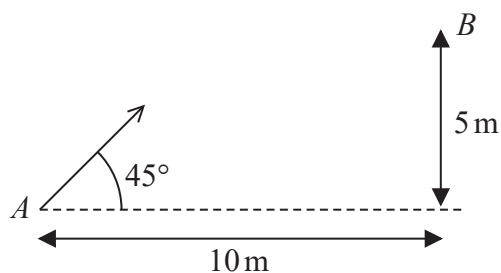


Figure 3

In a game, a small ball is thrown from a point A to a point B . The ball is thrown at 45° to the horizontal. The point B is 10 m horizontally from A and 5 m vertically from A , as shown in Figure 3.

In an initial model,

- the ball is modelled as a particle moving freely under gravity
- the initial speed of the ball is $U \text{ m s}^{-1}$
- $g = 9.8 \text{ m s}^{-2}$

Using this model,

- (a) find the value of U .

(6)

- (b) find the speed of the ball at B .

(3)

One limitation of this model is that the air resistance on the ball is ignored.

- (c) State one other limitation of this model.

(1)

In a refinement of the model,

- the ball is modelled as a particle
- air resistance on the ball is included
- the initial speed of the ball is $V \text{ m s}^{-1}$
- $g = 9.8 \text{ m s}^{-2}$

- (d) State how the value of V compares with the value of U , giving a reason for your answer.

(1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 11 marks)

TOTAL FOR MECHANICS IS 50 MARKS



Write your name here

Surname

Other names

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Candidate Number

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Mathematics

Advanced

Paper 3: Statistics and Mechanics

Sample Assessment Material for first teaching September 2017

Time: 2 hours

Paper Reference

9MA0/03

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer cross it out and put your new answer and any working out underneath.

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SECTION B: MECHANICS

Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

6. At time t seconds, where $t \geq 0$, a particle P moves so that its acceleration $\mathbf{a} \text{ m s}^{-2}$ is given by

$$\mathbf{a} = 5t\mathbf{i} - 15t^{\frac{1}{2}}\mathbf{j}$$

When $t = 0$, the velocity of P is $20\mathbf{i} \text{ m s}^{-1}$

Find the speed of P when $t = 4$

(6)

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Question 6 continued

(Total for Question 6 is 6 marks)

7. A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$.

A particle of mass m is placed on the plane and then projected up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is μ .

The particle moves up the plane with a constant deceleration of $\frac{4}{5}g$.

- (a) Find the value of μ .

(6)

The particle comes to rest at the point A on the plane.

- (b) Determine whether the particle will remain at A , carefully justifying your answer.

(2)

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8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors due east and due north respectively]

A radio controlled model boat is placed on the surface of a large pond.

The boat is modelled as a particle.

At time $t = 0$, the boat is at the fixed point O and is moving due north with speed 0.6 m s^{-1} .

Relative to O , the position vector of the boat at time t seconds is $\mathbf{r}$ metres.

At time $t = 15$, the velocity of the boat is $(10.5\mathbf{i} - 0.9\mathbf{j}) \text{ m s}^{-1}$.

The acceleration of the boat is constant.

- (a) Show that the acceleration of the boat is $(0.7\mathbf{i} - 0.1\mathbf{j}) \text{ m s}^{-2}$. (2)
- (b) Find $\mathbf{r}$ in terms of t . (2)
- (c) Find the value of t when the boat is north-east of O . (3)
- (d) Find the value of t when the boat is moving in a north-east direction. (3)

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Question 8 continued

(Total for Question 8 is 10 marks)

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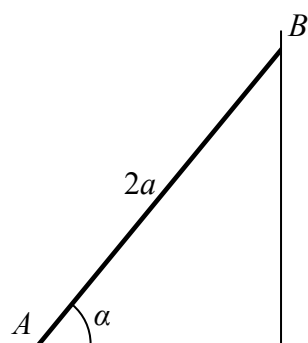


Figure 1

A uniform ladder AB , of length $2a$ and weight W , has its end A on rough horizontal ground.

The coefficient of friction between the ladder and the ground is $\frac{1}{4}$.

The end B of the ladder is resting against a smooth vertical wall, as shown in Figure 1.

A builder of weight $7W$ stands at the top of the ladder.

To stop the ladder from slipping, the builder's assistant applies a horizontal force of magnitude P to the ladder at A , towards the wall.

The force acts in a direction which is perpendicular to the wall.

The ladder rests in equilibrium in a vertical plane perpendicular to the wall and makes an angle α with the horizontal ground, where $\tan \alpha = \frac{5}{2}$.

The builder is modelled as a particle and the ladder is modelled as a uniform rod.

(a) Show that the reaction of the wall on the ladder at B has magnitude $3W$. (5)

(b) Find, in terms of W , the range of possible values of P for which the ladder remains in equilibrium. (5)

Often in practice, the builder's assistant will simply stand on the bottom of the ladder.

(c) Explain briefly how this helps to stop the ladder from slipping. (3)

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Question 9 continued

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Question 9 continued

(Total for Question 9 is 13 marks)

10.

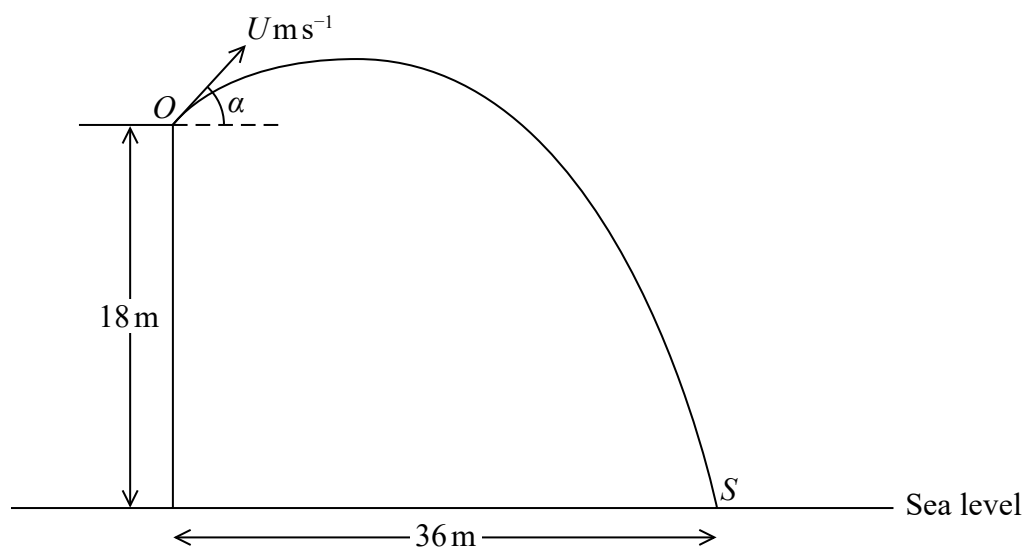


Figure 2

A boy throws a stone with speed $U \text{ m s}^{-1}$ from a point O at the top of a vertical cliff. The point O is 18 m above sea level.

The stone is thrown at an angle α above the horizontal, where $\tan \alpha = \frac{3}{4}$.

The stone hits the sea at the point S which is at a horizontal distance of 36 m from the foot of the cliff, as shown in Figure 2.

The stone is modelled as a particle moving freely under gravity with $g = 10 \text{ m s}^{-2}$

Find

- the value of U , (6)
- the speed of the stone when it is 10.8 m above sea level, giving your answer to 2 significant figures. (5)
- Suggest two improvements that could be made to the model. (2)

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Question 10 continued

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Question 10 continued

(Total for Question 10 is 13 marks)

TOTAL FOR SECTION B IS 50 MARKS
TOTAL FOR PAPER IS 100 MARKS

Write your name here

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Other names

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Centre Number

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Candidate Number

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Mathematics

Advanced Subsidiary

Paper 3: Statistics and Mechanics

Sample Assessment Material for first teaching September 2017

Time: 2 hours

Paper Reference

9MA0/03

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

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Information

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SECTION B: MECHANICS

Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

[In this question position vectors are given relative to a fixed origin O .]

6. A particle, P , moves with constant acceleration $(\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-2}$.

At time $t = 0$ seconds, the particle is at the point A with position vector $(2\mathbf{i} + 5\mathbf{j}) \text{ m}$ and is moving with velocity $\mathbf{u} \text{ m s}^{-1}$.

At time $t = 3$ seconds, P is at the point B with position vector $(-2.5\mathbf{i} + 8\mathbf{j}) \text{ m}$.

Find $\mathbf{u}$.

(4)

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 4 marks)

7. A particle, P , moves under the action of a single force in such a way that at time t seconds, where $t \geq 0$, its velocity $\mathbf{v}$ m s⁻¹ is given by

$$\mathbf{v} = (t^2 - 3t) \mathbf{i} - 12t \mathbf{j}$$

The mass of P is 0.5 kg.

Find the time at which the magnitude of the force acting on P is 6.5 N.

(7)

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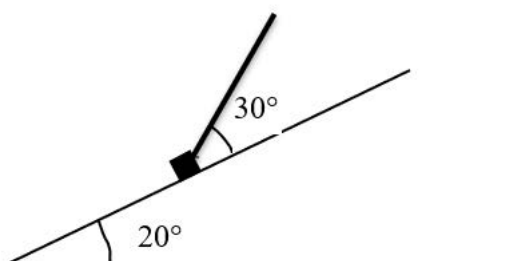


Figure 1

A small box of mass 3 kg moves on a rough plane which is inclined at an angle of 20° to the horizontal.

The box is pulled up a line of greatest slope of the plane using a rope which is attached to the box.

The rope makes an angle of 30° with the plane, as shown in Figure 1.

The rope lies in the vertical plane which contains a line of greatest slope of the plane.

The coefficient of friction between the box and the plane is 0.3.

The tension in the rope is 25 N.

The box is modelled as a particle, the rope is modelled as a light inextensible string and air resistance is ignored.

(a) Using the model, find the acceleration of the box.

(7)

(b) Suggest one improvement to the model that would make it more realistic.

(1)

The rope now breaks and the box slows down and comes to rest.

(c) Show that, after the box comes to rest, it immediately starts to move down the plane.

(3)

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 11 marks)

9.

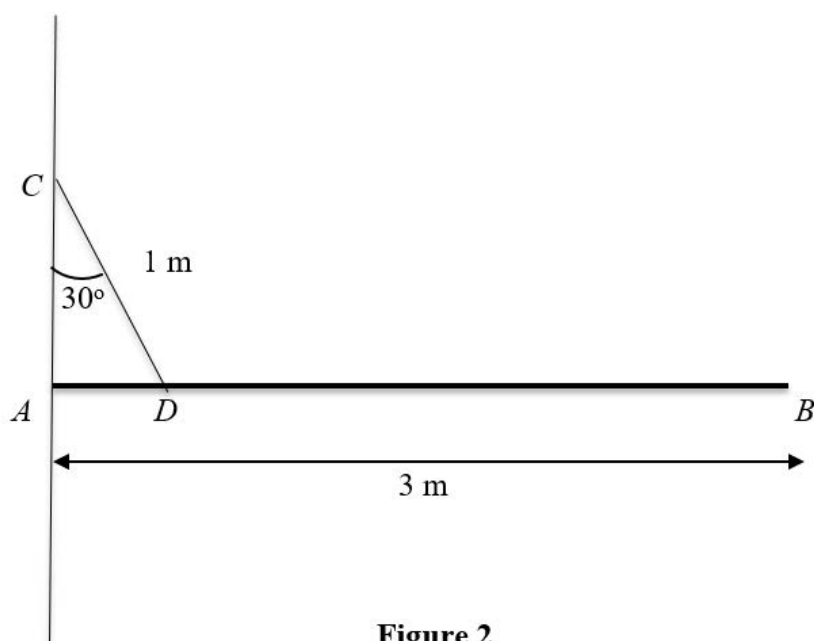


Figure 2

A beam AB , of mass 20 kg and length 3 m , is smoothly hinged to a vertical wall at one end A .

The beam is held in equilibrium in a horizontal position by a rope of length 1 m . One end of the rope is fixed to a point, C , on the wall which is vertically above A . The other end of the rope is fixed to the point D on the beam so that angle ACD is 30° , as shown in Figure 2.

The beam is modelled as a uniform rod and the rope is modelled as a light inextensible string.

Using the model, find

- the tension in the rope, (4)
- the direction of the force exerted by the wall on the beam at A . (6)
- If the rope were not modelled as being light, state how this would affect the tension in the rope, explaining your answer carefully. (2)

The rope is now removed and replaced by a longer rope which is still attached to the wall at C but is now attached to the beam at G , where G is the midpoint of AB . The beam AB remains in equilibrium in a horizontal position.

- Show that the force exerted by the wall on the beam at A now acts horizontally (2)

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(Total for Question 9 is 14 marks)

10.

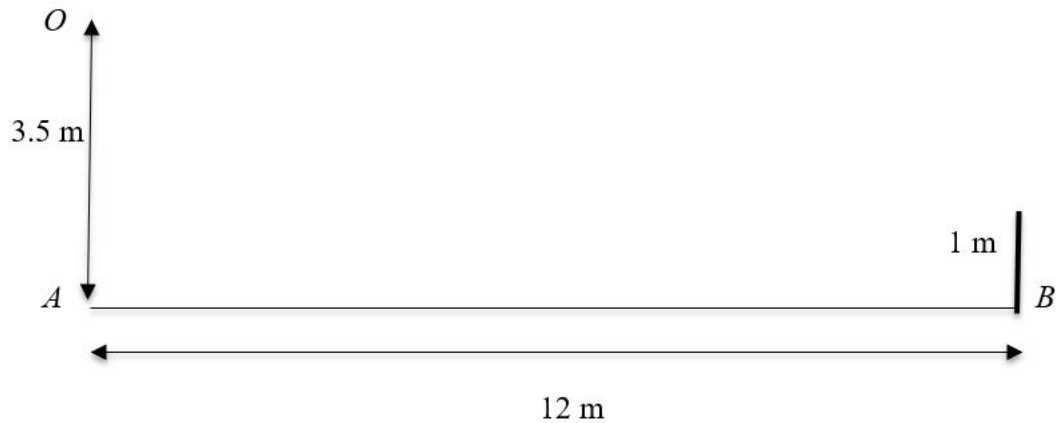


Figure 3

A tennis player serves a ball so as to pass over the net.

The ball is given an initial velocity of 45 m s^{-1} in a direction 10° below the horizontal.

The ball is struck at a point O which is 3.5 m vertically above the point A which is on horizontal ground.

The bottom of the net is the point B which is on the ground and $AB = 12 \text{ m}$.

The height of the net is 1 m , as shown in Figure 3.

The ball is modelled as a particle moving freely under gravity.

The ball passes over the net at a point which is vertically above B .

Using the model,

- find, in centimetres to 2 significant figures, the distance between the ball and the top of the net, as the ball passes over the net, (8)
- find, to 2 significant figures, the speed of the ball as it passes over the net. (4)
- State two limitations of the model that could affect the reliability of your answers. (2)

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 14 marks)

TOTAL FOR SECTION B IS 50 MARKS
TOTAL FOR PAPER IS 100 MARKS